

Total No. of Questions : 8]

SEAT No. :

PD4590

[Total No. of Pages : 2

[6404]-95

B.E (Computer Engineering)

DEEP LEARNING

(2019 Pattern) (410251) (Semester - VIII)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Make suitable assumption whenever necessary.

Q1) a) Draw and Explain CNN Architecture in detail. **[6]**

b) Explain working of Convolution Layer. **[6]**

c) Explain pooling layers and its type. **[6]**

OR

Q2) a) Explain all the features of pooling layer. **[6]**

b) Explain local response Normalization. **[6]**

c) Explain ReLU Layer in detail. **[6]**

Q3) a) Explain recursive neural network. **[6]**

b) Explain the LSTM in RNN. **[6]**

c) Explain in brief about working of RNN. **[5]**

OR

Q4) a) Difference between CNN Vs RNN. **[6]**

b) What are the challenges of long term dependencies? **[6]**

c) Explain Encoder-Decoder RNN model. **[5]**

P.T.O.

- Q5)** a) Explain Deep Generative Model. [6]
b) Explain Boltzmann Machine in details. [6]
c) Explain in brief GAN with an example. [6]

OR

- Q6)** a) Explain deep belief networks in detail. [6]
b) What is Generative Adversarial Network? Explain its component. [6]
c) Explain types of GAN. [6]

- Q7)** a) Explain Markov Decision Process. [6]
b) Explain deep reinforcement learning. [6]
c) What are the challenges of reinforcement learning? [5]

OR

- Q8)** a) Explain the process of Deep Q-learning. [6]
b) Explain reinforcement learning for Tic-Tac-Toe game. [6]
c) Explain Dynamic programming algorithm for reinforcement learning. [5]

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